

Economic Audit Report

For  **SUPERLEND**

Oct 2024



Superlend Economic Audit | Etherlink

Chainrisk

October 2024

Abstract

Superlend is an upcoming decentralized lending protocol that enables secure and efficient borrowing and lending of cryptocurrencies. It utilizes smart contracts and dynamic interest rates based on supply and demand to facilitate transactions. The protocol allows users to supply different crypto assets and accrue interest, while borrowers can avail themselves of loans secured by collateralized assets. Our collaboration with Superlend focuses on harnessing rigorous data analysis through a battery of statistical tools to optimize the protocol's critical risk parameters. This report delineates a comprehensive methodology for calculating the protocol's key risk metrics. This optimization framework is pivotal for mitigating market risks and enhancing the protocol's stability.

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2 About Chainrisk

Chainrisk is a Security SaaS company building a unified simulation platform that allows teams to test protocols efficiently while understanding how they will react to adversarial market environments. The backbone of our technology is a rigorous statistical analysis and modeling of historical data, and a cloud-based, agent- and scenario-based simulation engine. This allows us to create specific market scenarios to stress test the protocol, calculate Value at Risk (VaR), and optimize risk parameters.

3 Background

3.1 Overview

Superlend is a comprehensive DeFi lending and borrowing aggregator that integrates multiple money markets across various blockchains. Starting with the Etherlink market, Superlend aggregates liquidity and opportunities from over 30 leading DeFi protocols across platforms like Ethereum, Arbitrum, Optimism, and Base, offering users the highest yields and competitive borrowing rates. Designed to simplify and enhance the DeFi experience, Superlend provides access to a wide array of markets through an open-source platform, allowing users and developers to interact with the protocol via a user interface, API, or directly through smart contracts, fostering a robust and versatile ecosystem.

3.2 Scope of Audit

The scope of the audit is the Etherlink chain, specifically focusing on the market for the following assets - Wrapped Bitcoin (WBTC), Wrapped Ether (WETH), USD Coin (USDC), Tether (USDT), and Tezos (XTZ). It is to be noted that in the rest of the document, Wrapped Ether (WETH) and Wrapped Bitcoin (WBTC) will be interchangeably used with Ethereum (ETH) and Bitcoin (BTC), respectively. This can be done as WETH and WBTC are tokenized versions of ETH and BTC in DeFi markets. We will conduct a thorough examination of both the collateral and borrowable assets within this market. Our evaluation will include extreme value analysis of historical data for the protocol to assess its performance and recommend optimal risk parameters for the Etherlink chain to ensure market stability and security.

3.3 Goals of the Analysis

In this paper, we present an in-depth market analysis of the Superlend protocol on the Etherlink chain, where we calculated risk parameters by examining various critical factors such as liquidity, volatility, and supply and demand curves.

By applying our market risk calculation framework to the Superlend market, we recommend optimal parameters that balance capital efficiency and risk, ultimately enhancing the stability and sustainability of the Superlend market on Etherlink.

4 Protocol Overview

4.1 Collateral and Borrowing

In Superlend, borrowing is an over-collateralized process where users must supply collateral assets to borrow the base asset. This over-collateralization serves as a safeguard against market volatility and potential insolvency, ensuring that the protocol maintains a robust collateralization ratio. For Example, Consider a scenario where a user intends to borrow USDC by using ETH as collateral.

Collateral Asset: ETH

Base Asset: USDC

Borrow Collateral Factor for ETH: 73%

User Supplies: 3 ETH, valued at \$8,100 (assuming an ETH price of \$2,700).

Calculation of Borrowing Capacity: With the Borrow Collateral Factor of 73%, the user can borrow up to 73% of the value of their supplied ETH.

To calculate the borrowing capacity:

Borrowing Capacity = Value of collateral Supplied $\times$ Collateral Factor

Borrowing Capacity = $\$8100 \times 0.73 = \5913

With the ETH price at \$2,700 and a Borrow Collateral Factor of 73%, the user can borrow a maximum of \$5,913 worth of USDC against their 3 ETH collateral.

4.2 Interest Rate Curve

4.2.1 Borrow Interest Rate

Mitigating liquidity risk through the borrow interest rate model Superlend's interest rate algorithm is calibrated to manage liquidity risk and optimize utilization. The borrow interest rates are derived from the Utilisation Rate U .

U is an indicator of the availability of capital within the pool. The interest rate model manages liquidity risk in the protocol through user incentives to support liquidity:

- **When capital is available:** low interest rates to encourage borrowing.
- **When capital is scarce:** high interest rates encourage repayments of debt and additional supplying.

4.2.2 Interest Rate Model

Liquidity risk materializes when utilization is high, and this becomes more problematic as U gets closer to 100%. To tailor the model to this constraint, the interest rate curve is split into two parts around an optimal utilization rate $U_{optimal}$. Before $U_{optimal}$, the slope is small, after it begins rising sharply.

The interest rate R_t follows the model as stated below:

$$\begin{aligned} \text{if } U \leq U_{optimal} : R_t &= R_0 + \left(\frac{U_t}{U_{optimal}} \right) R_{slope1}, \\ \text{if } U > U_{optimal} : R_t &= R_0 + R_{slope1} + \left(\frac{U_t - U_{optimal}}{1 - U_{optimal}} \right) R_{slope2}, \end{aligned}$$

where R_{slope1} and R_{slope2} are two variables for rate slopes. For different strategies, the values of these slopes vary and are assumed respectively.

4.2.3 Supply Rate

The borrow interest rates paid are distributed as yield for aToken holders who have supplied to the protocol, excluding a share of yields sent to the ecosystem reserve defined by the reserve factor. This interest rate is generated on the asset that is borrowed out then shared among all the liquidity providers. The supply APY, D_t , is :

$$S_t = U_t (SB_t S_t + VB_t V_t) (1 - R_t),$$

where,

- U_t : Utilization Rate,
- SB_t : the Share of Stable Borrows,
- S_t : the average stable rate,
- VB_t : the share of variable borrows,
- V_t : the variable rate, and
- R_t : the reserve factor.

4.3 Liquidation Mechanism

Liquidation is a process that occurs when a borrower's health factor goes below 1 due to their collateral value not properly covering their loan/debt value.

This might happen when the collateral decreases in value or the borrowed debt increases in value against each other. This collateral vs loan value ratio is shown in the health factor.

In a liquidation, up to 50% of a borrower's debt is repaid and that value plus liquidation fee is taken from the collateral available, so after a liquidation that amount liquidated from your debt is repaid.

To illustrate the liquidation process in Superlend, consider the following scenario:

Example 1

Bob deposits 10 ETH and borrows 5 ETH worth of DAI.

If Bob's Health Factor drops below 1 his loan will be eligible for liquidation.

A liquidator can repay up to 50% of a single borrowed amount = 2.5 ETH worth of DAI.

In return, the liquidator can claim a single collateral which is ETH (5% bonus).

The liquidator claims $2.5 + 0.125$ ETH for repaying 2.5 ETH worth of DAI.

Example 2

Bob deposits 5 ETH and 4 ETH worth of YFI, and borrows 5 ETH worth of DAI.

If Bob's Health Factor drops below 1 his loan will be eligible for liquidation.

A liquidator can repay up to 50% of a single borrowed amount = 2.5 ETH worth of DAI.

In return, the liquidator can claim a single collateral, as the liquidation bonus is higher for YFI (15%) than ETH (5%) the liquidator chooses to claim

YFI.

The liquidator claims $2.5 + 0.375$ ETH worth of YFI for repaying 2.5 ETH worth of DAI.

To avoid liquidation you can raise your health factor by depositing more collateral assets or repaying part of your loan. By default, repayments increase your health factor more than deposits. Also, it's important to monitor your health factor and keep it high to avoid liquidation. Keeping your health factor over 2, for example, gives you more of a margin to avoid a liquidation.

5 Risk Parameters Definitions

5.1 Supply Caps

Supply caps limit the maximum amount of a particular asset that can be supplied to a protocol. For instance, if a protocol sets a supply cap of 1,000 tokens for a newly launched cryptocurrency, it prevents more than 1,000 tokens from being added to the system. This helps manage the protocol's risk, especially for highly volatile assets and protects against potential exploits like infinite minting. The supply cap is usually based on factors such as the on-chain liquidity of the asset and the total volume of collateral in the pool.

5.2 Borrow Caps

Borrow caps set the maximum amount of an asset that can be borrowed from a protocol. For example, if a protocol establishes a borrow cap of 500 tokens for a certain cryptocurrency, it means no more than 500 tokens can be

borrowed at any time. This measure helps prevent issues like traditional or flash loans of an asset that might be vulnerable to price manipulation, which could lead to the protocol becoming insolvent. The borrow cap is generally determined by factors such as the asset's on-chain liquidity and the total volume of borrowed assets in the pool.

5.3 Isolation Mode

Isolation mode is a feature designed to minimize systemic risk when listing riskier assets. For example, if a protocol lists a highly volatile cryptocurrency, it can activate isolation mode to limit the asset's interaction with the broader pool. In this mode, the asset can only be used as collateral for borrowing specific, isolated stablecoins and cannot be combined with other assets as collateral. This means that a user could only use this single risky asset to borrow a designated stablecoin, reducing the risk to the overall system.

5.4 Siloed Mode

Siloed Mode is a feature used to reduce the risk of insolvency when listing new assets with potentially manipulatable oracles. For instance, if a protocol lists an illiquid Uni pair, where a single trade could drastically impact the price, it might activate Siloed Mode to mitigate this risk. In Siloed Mode on the Superlend protocol, a user borrowing a siloed asset, like this illiquid pair, is restricted to borrowing only that asset and cannot borrow any other assets simultaneously. This helps protect the protocol from potential exploits that could arise from price manipulation.

5.5 eMode

Efficient Mode (or "eMode") enables assets with correlated prices, such as DAI, USDC, and USDT, to be grouped into the same eMode category. This grouping improves capital efficiency by allowing higher loan-to-value (LTV) ratios when both the borrowed asset and the collateral are in the same eMode category. For example, if DAI, USDC, and USDT are classified under a "Stablecoins" eMode category, users can borrow more against these assets compared to assets outside this category.

Currently, Superlend's markets have one defined eMode category—Stablecoins, category 1. However, Superlend permits `RISK_ADMIN` and `POOL_ADMIN`, appointed by Superlend Governance, to create and configure up to 255 eMode categories. Each eMode category can include the following risk management settings:

- LTV (loan-to-value ratio),
- Liquidation Threshold,
- Liquidation Bonus, and
- Optional custom price oracle.

5.6 Loan to Value

The Loan-to-Value (LTV) ratio determines the maximum amount of assets that can be borrowed based on the value of the collateral. It is expressed as a percentage. For instance, with an LTV of 75%, a borrower can take out 0.75 ETH worth of a currency for every 1 ETH of collateral they provide. As

market conditions change, the LTV ratio will adjust accordingly. For example, if the value of ETH drops, the amount of currency a borrower can access might decrease to maintain the LTV ratio.

5.7 Liquidation Threshold

The liquidation threshold is the percentage at which a position is considered undercollateralized and may be subject to liquidation. For example, if the liquidation threshold is set at 80%, this means that if the value of the collateral falls below 80% of its original value, the position becomes undercollateralized and could be liquidated.

The difference between the Loan-to-Value (LTV) ratio and the liquidation threshold acts as a safety buffer for borrowers.

For each wallet, the liquidation threshold is calculated as the weighted average of the thresholds for the collateral assets and their respective values. The formula is:

$$Liquidation\ Threshold = \frac{\sum_i Collateral_i\ in\ ETH \times Liquidation\ Threshold_i}{Total\ Collateral\ in\ ETH}$$

For example, if a wallet holds 1 ETH of Asset A with a liquidation threshold of 75% and 2 ETH of Asset B with a liquidation threshold of 80%, the overall liquidation threshold for the wallet would be a weighted average based on the values and thresholds of these assets.

5.7.1 Liquidation Penalty

The liquidation penalty is a fee applied to the price of the collateral assets when they are sold by liquidators during the liquidation of a loan that has fallen below the liquidation threshold.

For example, if a borrower's position is liquidated and the liquidation penalty (called the liquidation bonus in Aave) is set at 5%, the liquidator would receive 5% more collateral than the debt they repaid on behalf of the borrower. So, if the liquidator repays enough debt to liquidate 100 ETH worth of collateral, they would receive 105 ETH in return, with the extra 5 ETH representing the liquidation bonus as an incentive for liquidating the position.

5.7.2 Liquidation Factor

The liquidation factor determines the portion of the liquidation penalty that is allocated to a collector contract from the ecosystem's treasury.

For example, if a liquidation penalty of 5% is imposed and the liquidation factor specifies that 2% of this penalty goes to the collector contract, then for a liquidation involving assets worth 100 ETH, 2 ETH (which is 2% of the 5 ETH penalty) will be directed to the collector contract, while the remaining 3 ETH goes elsewhere as per the protocol's rules.

5.7.3 Reserve Factor

The reserve factor designates a portion of the protocol's interest earnings to a collector contract from the ecosystem's treasury.

Superlend manages its solvency risk through the Safety Module, which is funded by incentives from the ecosystem reserve. Consequently, the reserve factor is a risk premium that adjusts according to the asset's overall risk. For example, stablecoins, which are less risky, have a lower reserve factor, while more volatile assets, which carry higher risk, have a higher reserve factor. So, if the reserve factor for stablecoins is set at 1% and for volatile assets at 5%, this means that for every 100 ETH earned in interest, 1 ETH would go to the collector contract for stablecoins. In comparison, 5 ETH would be allocated for volatile assets.

6 Statistical Analysis

6.1 Assumptions

- **Probability Distribution** - We assume the probability distribution function of the volatility estimate (say, log returns, price difference ratio, others) for each asset will remain consistent over the upcoming time duration. This steadiness lets us use past data as a solid foundation to analyze future risks and make choices.
- **Unbiasedness** - The volatility estimate is unbiased.
- **Consistency** - The volatility estimate is also consistent.

6.2 Introduction

Over time, percentiles have been increasingly used as an alternative to mean-based indicators in various scenarios [1]. They are particularly valuable in

situations where simple mean-based statistics fail to capture the full picture, especially in skewed distributions or when outliers are present. Theoretically, a percentile represents the value below which a certain percentage of the data points fall, providing a more nuanced understanding of the distribution's shape and spread.

Now, it is necessary that we are able to visualize the percentiles rather than only just delve into the mathematics. For that purpose, we focus on pictorial representations of the data. Boxplots have become a favored tool for visualizing data distributions, especially when a quick summary of the data's central tendency, spread, and potential outliers is needed [3]. Unlike mean-based representations, boxplots do not assume symmetry in the data and can reveal insights into the distribution's overall structure, such as skewness and the presence of extreme values. This makes them particularly useful in identifying patterns that may be hidden when relying solely on mean-based indicators.

The Mean Absolute Deviation (MAD) has also emerged as a robust alternative to mean-based measures of variability [2]. While standard deviation is more common, MAD is less sensitive to extreme values, making it more reliable in cases where outliers could distort the interpretation of data spread. By calculating the average absolute difference between each data point and the mean, MAD provides a clearer sense of typical deviations, offering a more accurate reflection of a dataset's volatility or consistency, especially in non-normal distributions.

6.3 Data Description

The datasets used in this analysis contain the price movements of five different assets: USDT, USDC, WBTC, WETH, and XTZ. The statistical analysis and recommendations provided in this report are based on the data available on the Etherlink Chain.

Each dataset includes two key columns: timestamp, which records the specific times at which prices were observed, and price, which records the corresponding asset prices in respective units. By analyzing these datasets using a battery of statistical analysis, we aim to understand these assets' price trends and patterns over the given period, which might help us make respective recommendations.

The price movements of the assets (here for Wrapped Ether (WETH), Wrapped Bitcoin (WBTC), and Tezos (XTZ)) are visualized using line plots. Each plot illustrates the fluctuations in price over time for each asset, offering a preliminary understanding of the trends and potential volatility in the corresponding market.

It is to be noted, from Figures 1, 2, and 3 we can interpret the following :

- The plots indicate different behaviors and trends in the prices of these cryptocurrencies.
- These trends suggest varying levels of market confidence and investor behavior across different cryptocurrency pairs and networks.
- The above line plots show no statistically significant correlation amongst price movements of the assets.

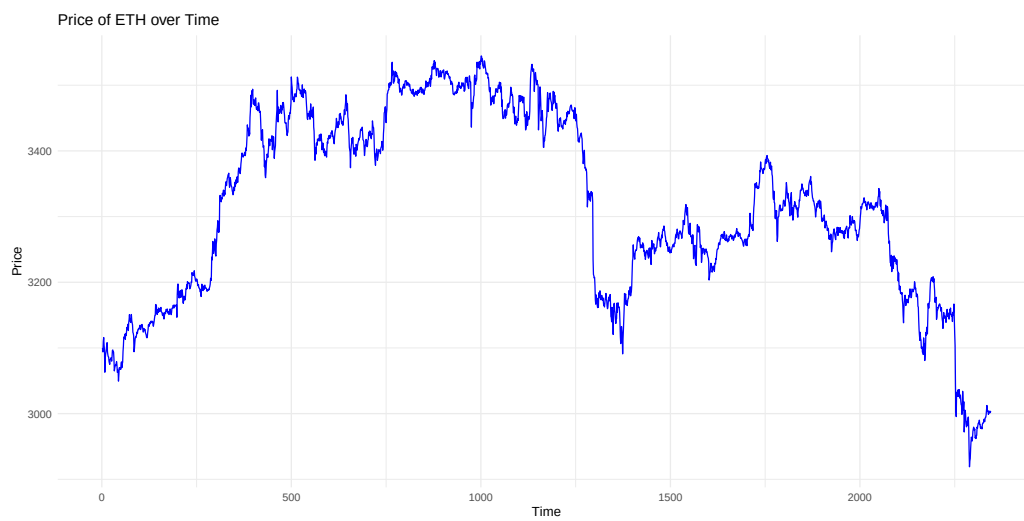


Figure 1: Wrapped Ether Price Movement

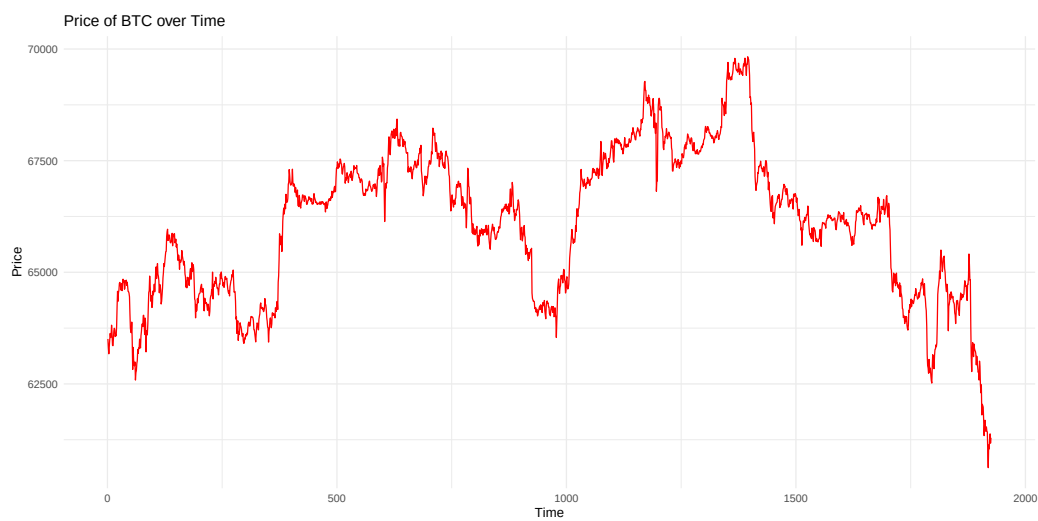


Figure 2: Wrapped Bitcoin Price Movement



Figure 3: Tezos Price Movement

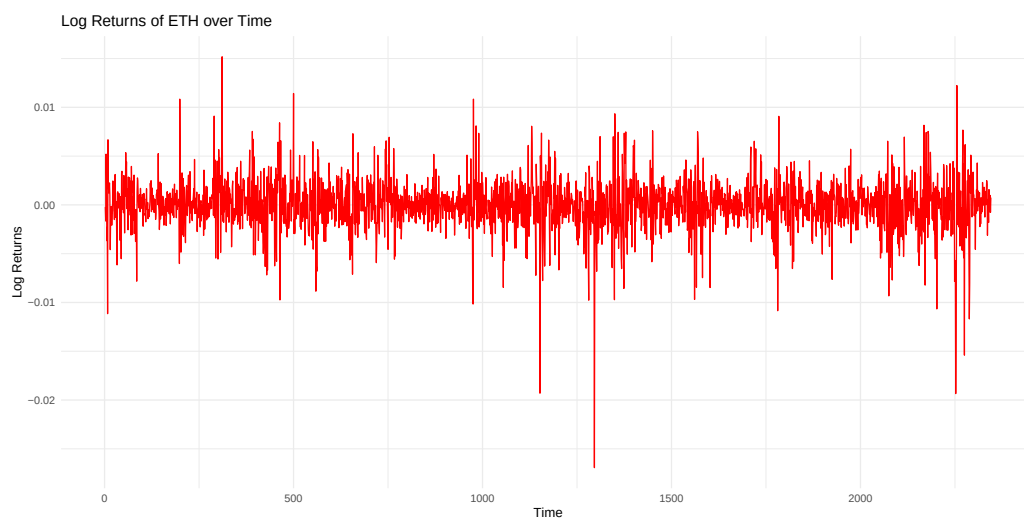


Figure 4: Historical Volatility Estimate for Wrapped Ether

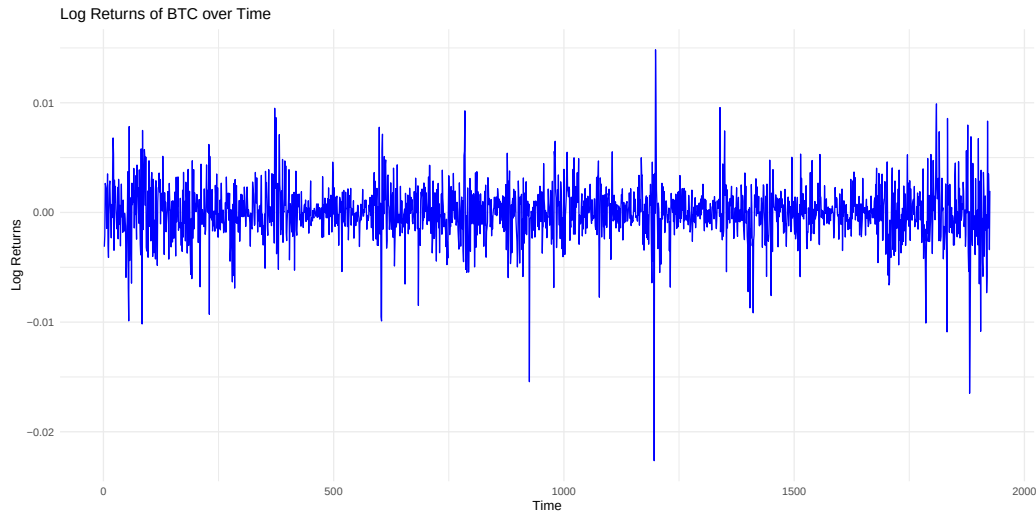


Figure 5: Historical Volatility Estimate for Wrapped Bitcoin

Figure 4 of Wrapped Ether appears to have a similar range as Bitcoin, but the fluctuations seem to have more pronounced spikes, particularly early on. This might suggest higher volatility during certain periods.

Figure 5 shows how the estimate of volatility (here, log returns) of Bitcoin fluctuates over time, with values mostly oscillating between -0.01 and 0.01 , indicating relatively stable price movements with occasional spikes or drops.

Figure 6 shows the volatility for Tezos asset. Here, we observe that the volatility has a wider range between -0.02 and 0.02 , indicating potentially higher volatility compared to Bitcoin and Ethereum. The spikes are visible but more evenly spread out across the time series. The price drop has been observed as low as near -0.03 , which indicates the high volatility of the Tezos asset as of the current time.

In general, these plots are useful for identifying periods of high volatility or stability in the respective cryptocurrencies and can be used for further

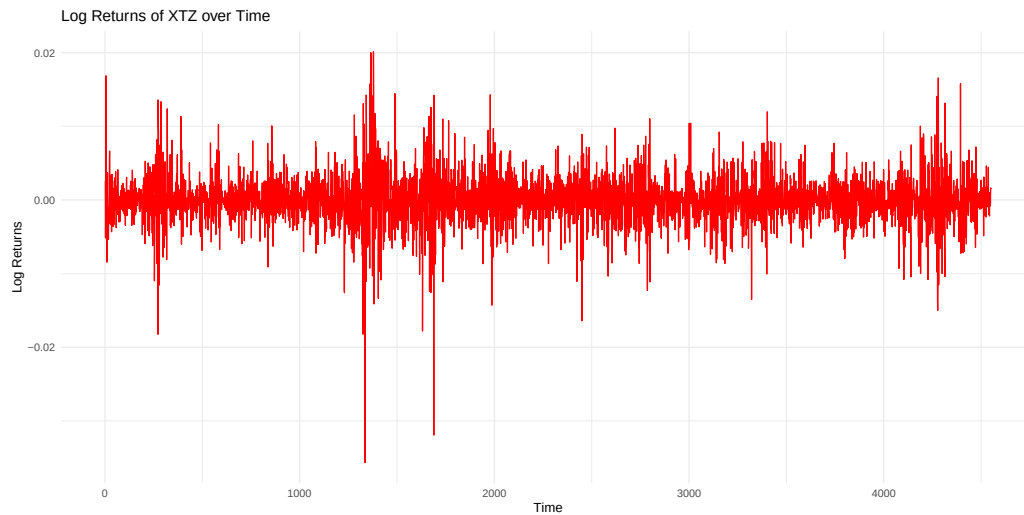


Figure 6: Historical Volatility Estimate for Tezos

analysis, such as calculating risk measures or developing trading strategies.

6.4 Analysis and Results

6.4.1 Conservative Method (Absolute Deviation)

In a conservative method, we try to provide a safe strategy option. Thus, the conservative method needs to center on the overall variability in returns, which we measure through the absolute deviation of the volatility estimate. This approach provides a wider view, namely the extreme scenario of the typical fluctuations one might expect over any time period.

With USDT as a Borrow Asset

- ETH : 43.62129%
- BTC : 39.58943%
- XTZ : 46.24707%

- USDC : 0%

With USDC as a Borrow Asset

- ETH : 43.62129%
- BTC : 39.58943%
- XTZ : 46.24707%
- USDT : 0%

With ETH as a Borrow Asset

- BTC : 47.60412%
- XTZ : 44.53229%
- USDC : 45.77497%
- USDT : 45.77497%

With BTC as a Borrow Asset

- ETH : 47.60412%
- XTZ : 47.55143%
- USDC : 39.58151%
- USDT : 39.58151%

With XTZ as a Borrow Asset

- ETH : 44.53229%

- BTC : 47.55143%
- USDC : 45.46004%
- USDT : 45.46004%

Though the line plot of prices shows the expected annual fluctuations in the log returns for each asset, we used this method to give a careful estimate of the risk tied to each asset. It puts more weight on the overall spread of returns instead of specific percentile-based limits to give us a robust understanding of the price changes.

6.4.2 Aggressive Method (Percentile Values)

In the aggressive method, we ignore the extreme drop of prices (extreme values of log returns) and take a specific percentile value (which will capture negative but not extreme ones) of log returns. We assume consistency in the probability distribution function of the log returns. The percentile value has been further standardized using a scaling factor.

When USDT is the Borrow Asset

35th Percentile:

- ETH : -15.12872%
- BTC : -14.47886%
- XTZ : -19.53636%
- USDC : 0%

When USDC is the Borrow Asset

35th Percentile:

- ETH : -15.12872%
- BTC : -14.47886%
- XTZ : -19.53636%
- USDT : 0%

When ETH is the Borrow Asset

35th Percentile:

- BTC : -16.47471%
- XTZ : -17.32421%
- USDC : -16.42478%
- USDT : -16.42478%

When BTC is the Borrow Asset

35th Percentile:

- ETH : -19.44498%
- XTZ : -18.57283%
- USDC : -14.18493%
- USDT : -14.18493%

When XTZ is the Borrow Asset

35th Percentile:

- ETH : -15.27339%
- BTC : -15.98235%
- USDC : -18.95298%
- USDT : -18.95298%

We are only concerned with percentile values so that we can capture the possible negative returns. It is to be noted that future negative returns will not cross this threshold of earlier given values with a probability of 0.65.

Boxplots of log returns of all three assets trace quite nicely their distribution. The central box here shows the IQR, or the middle 50% of the data, and in that box, a line indicating the median is here very close to zero, meaning that the normal log return is close to zero. Whiskers extend showing the range within 1.5 times the IQR, and circles outside these whiskers highlight outliers—returns significantly off the norm. Taken altogether, this set of plots suggests that while most returns are stationary, there are periods in which asset prices undergo a wild swing.

7 Risk Parameter Testing Methodology

The risk parameters mitigate the market risks of the assets supported by the Superlend protocol. Each borrow is based on an over-collateralization with a different asset that may be subject to volatility. Sufficient margins

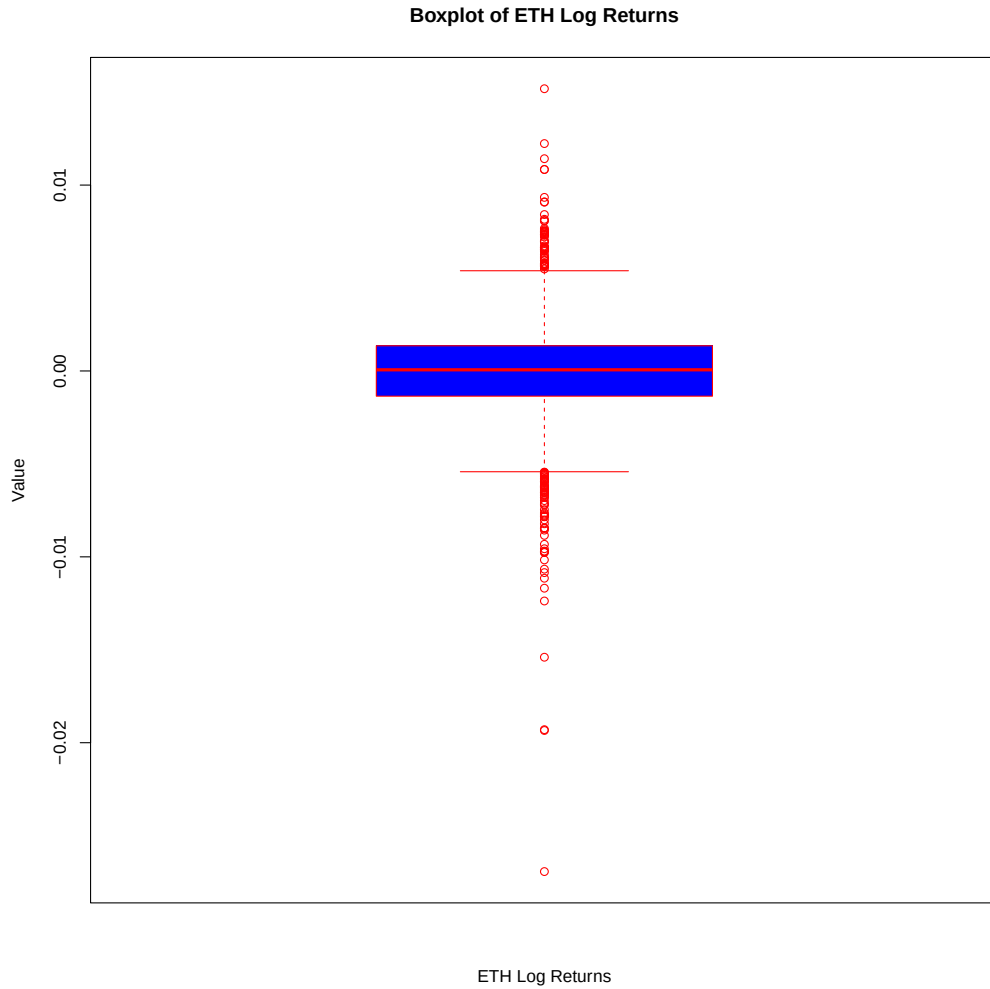


Figure 7: Box plot for Volatility estimate of Ethereum

and incentives are needed for the position to remain collateralized in the event of adverse market conditions. If the value of the collateral falls below a predetermined threshold, a portion of it will be auctioned as a liquidation bonus to repay a portion of the debt position and keep the ongoing borrow collateralized.

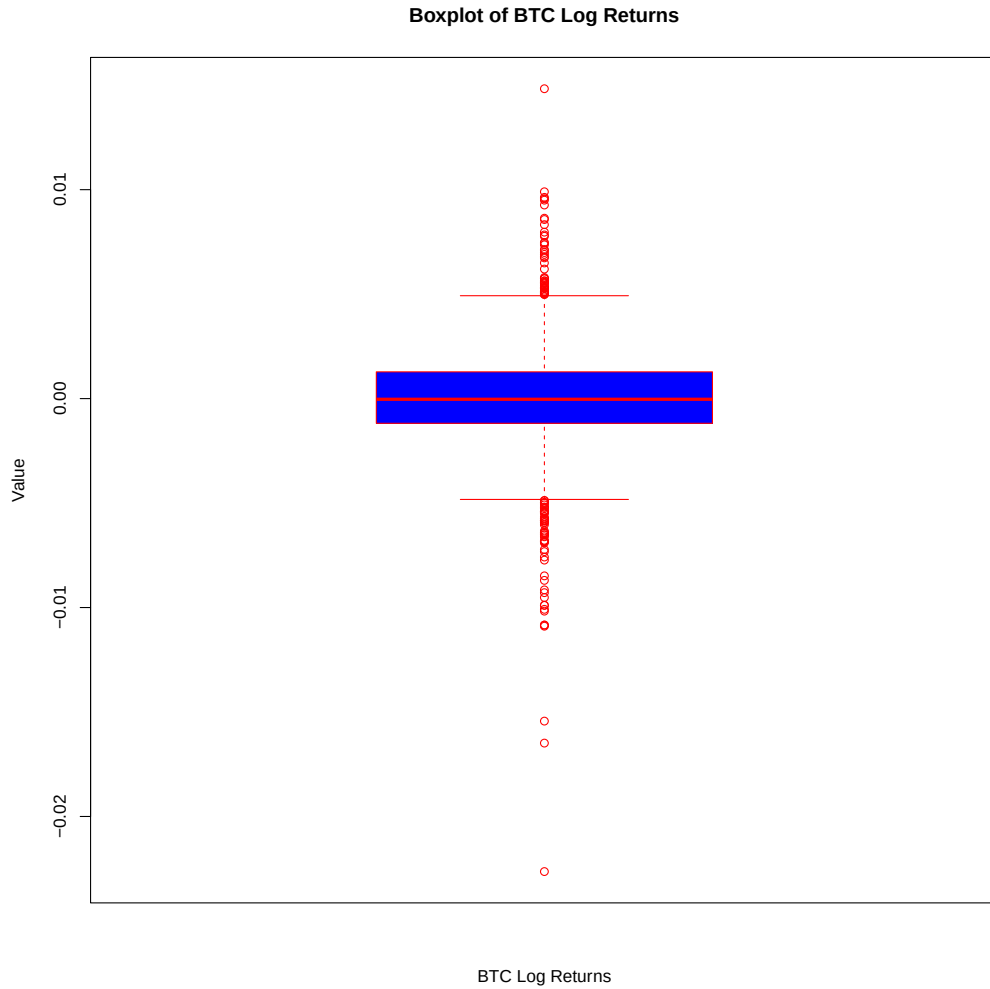


Figure 8: Box plot for Volatility estimate of Bitcoin

Market risks can be mitigated through different risk parameters, which define collateralization and liquidation rules.

These parameters are calibrated on a per-asset basis to account for the specific risks identified.

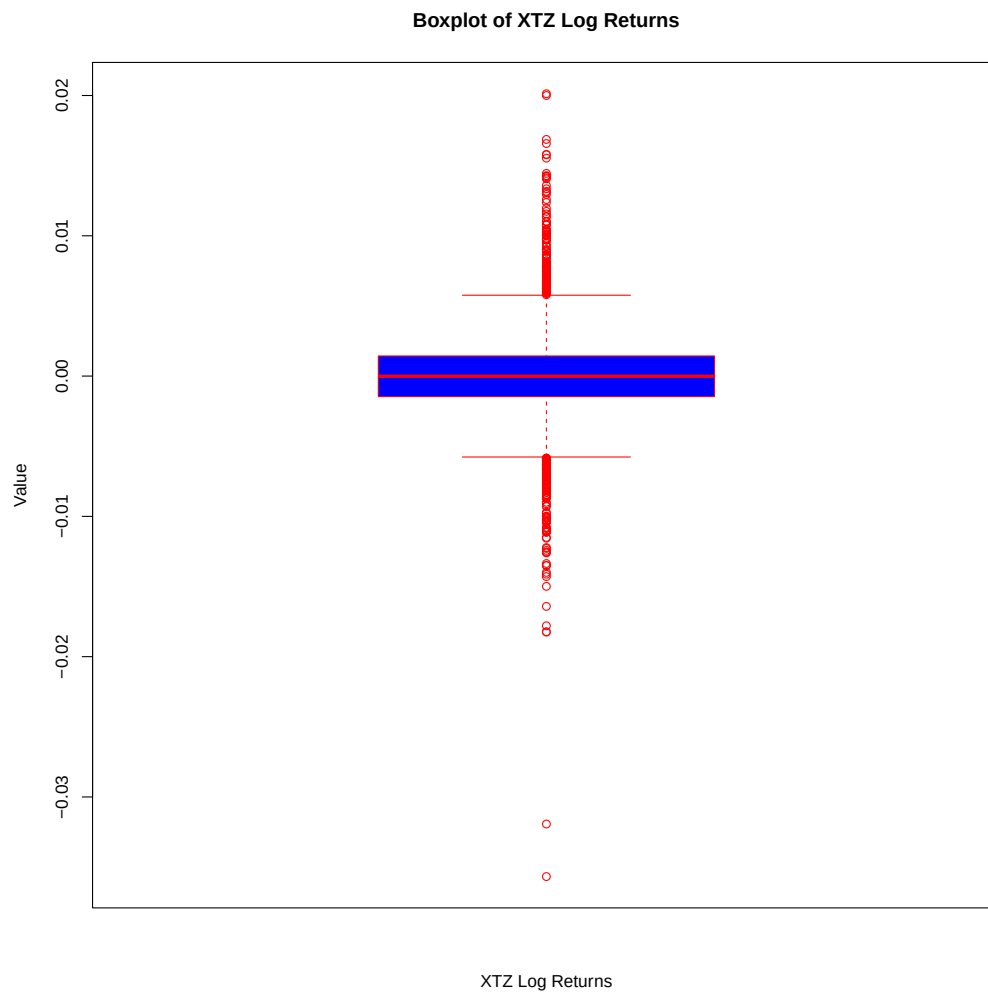


Figure 9: Box plot for Volatility estimate of Tezos

7.1 Supply Caps

Our conservative recommendation framework is summarized below.

- Stablecoin: 40% of the circulating token supply on-chain.
- Non-Stablecoin: 30% of the circulating token supply on-chain.

Our aggressive recommendation framework is summarized below.

- Stablecoin: 60% of the circulating token supply on-chain.
- Non-stablecoin: 50% of the circulating token supply on-chain.

7.2 Borrow Caps

The borrow cap is the maximum amount of that collateral asset that can be borrowed against. This parameter is crucial in preventing excessive borrowing that could lead to systemic risks, like insolvency during market downturns or liquidity crises.

The borrow cap for each asset within the Superlend protocol has been meticulously calibrated to ensure both security and liquidity.

Borrow Cap Setting: For assets like WBTC, WETH, and XTZ, the borrow cap has been set at 60% of the established supply cap. For stablecoins like USDT and USDC, the borrow cap has been set at 90% of the supply cap. Stablecoins, by their nature, are less volatile and offer a higher degree of price stability compared to assets.

This cap ensures a balanced and secure borrowing environment, minimizing the risk of liquidity issues. This cap is subject to adjustment in the future

based on the assessed risk associated with specific assets. This approach ensures that only a portion of the supplied collateral is leveraged for borrowing, thereby mitigating systemic risks such as excessive borrowing, market-induced insolvencies, and potential liquidity crises.

The borrow cap can't exceed the supply cap of the asset.

7.3 Isolation Mode

Isolation mode can effectively limit systemic risk when listing riskier assets. This mode restricts an asset to borrowing only isolated stablecoins and allows only one isolated asset to be used as collateral at any given time.

Given the current market conditions of Etherlink Testnet, which are characterized by low trading volumes, we recommend against the implementation of isolation mode at this time. Isolation mode is typically employed to mitigate systemic risk by restricting riskier assets to borrowing only specific stablecoins and allowing only one isolated asset to be used as collateral. However, in a market with limited activity, such as Etherlink, the potential for systemic risk is inherently low, and introducing a debt ceiling could unnecessarily constrain growth as the overall market conditions do not pose a significant systemic risk. Therefore, maintaining the current structure without isolation mode is advisable to support market flexibility and development.

7.4 Siloed Mode

Siloed Mode is a feature used to reduce the risk of insolvency when listing new assets with potentially manipulatable oracles. For instance, if a protocol lists an illiquid Uni pair, where a single trade could drastically impact the

price, it might activate Siloed Mode to mitigate this risk. In Siloed Mode on the Superlend protocol, a user borrowing a siloed asset, like this illiquid pair, is restricted to borrowing only that asset and cannot borrow any other assets simultaneously. This helps protect the protocol from potential exploits that could arise from price manipulation.

7.5 eMode

Efficient Mode ("eMode") enhances capital efficiency by grouping price-correlated assets, such as DAI, USDC, and USDT, into the same category. This allows for higher Loan-to-Value (LTV) ratios for both borrowed and collateral assets within that category, making it more attractive for users to maximize their borrowing potential. eMode will be enabled in the Etherlink market to encourage user participation by offering the advantage of higher borrowing capacity, thereby attracting more users to the platform and fostering greater market activity.

In the Etherlink market, USDC and USDT have been chosen for Efficient Mode ("eMode") due to their strong price stability and correlation. We're setting a Loan-to-Value (LTV) ratio of 97% and a Liquidation Threshold (LT) of 98.5% for these assets.

With a 97% LTV, users can borrow up to nearly the full value of their collateral, offering them greater borrowing power while maintaining a low risk of liquidation. The 98.5% Liquidation Threshold is set to provide a strong safety margin, ensuring the stability of user positions even during market volatility. This setup is aimed at encouraging more users to take advantage of Etherlink's platform by maximizing the potential of their stablecoin assets.

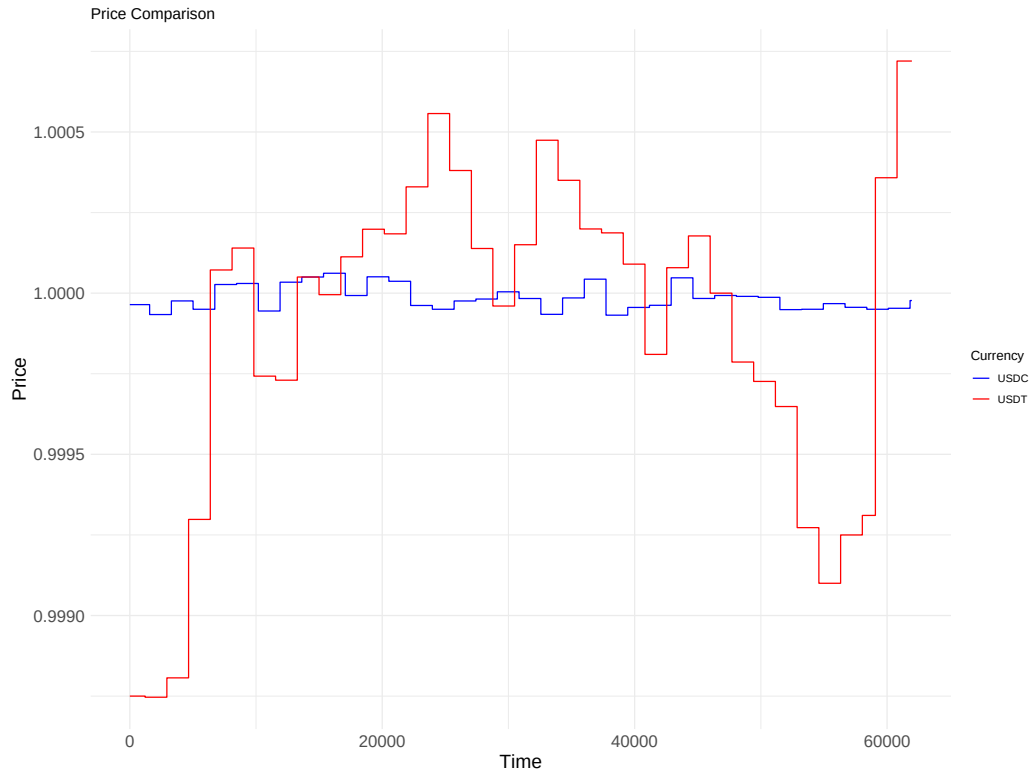


Figure 10: Price comparison of USD Coin (USDC) and Tether (USDT)

To illustrate the benefits of this approach, a comparison graph has been added, highlighting how these parameters offer a significant advantage over standard lending practices. This graph clearly demonstrates how eMode can enhance borrowing capacity, making it an appealing option for those looking to fully utilize their assets in the market.

Figure 10 gives an overall idea about the comparative price movement of USDC, and USDT assets for a time interval, and hence will give an idea about the values set for the risk parameters.

7.6 Loan to Value

The Loan to Value (LTV) ratio defines the maximum amount of assets that can be borrowed using a specific collateral.

In a lending protocol like Superlend, the Loan-to-Value (LTV) ratio is crucial for determining how much can be borrowed against collateral. Here's an analysis of each asset, recommending one as the borrowing asset and the others as collateral.

7.6.1 WBTC (Wrapped Bitcoin)

- Borrowing Asset: WBTC
- Collateral: WETH, USDC, USDT, XTZ

Rationale: WBTC is a stable asset backed by Bitcoin, making it a relatively safe borrowing choice. When using WETH, USDC, USDT, and XTZ as collateral, the LTV should be conservative to mitigate volatility risks associated with Ethereum and altcoins.

7.6.2 WETH (Wrapped Ether)

- Borrowing Asset: WETH
- Collateral: WBTC, USDC, USDT, XTZ

Rationale: WETH is often used for trading and can be a good borrowing asset due to its liquidity. However, it is more volatile than WBTC.

7.6.3 USDC (USD Coin)

- Borrowing Asset: USDC
- Collateral: WBTC, WETH, USDT, XTZ

Rationale: USDC is a stablecoin, making it an ideal borrowing asset for those seeking to minimize risk. Collateralizing with WBTC, WETH, USDT, and XTZ allows for a more stable backing, with an LTV of up to 80% being reasonable due to the stability of USDC.

7.6.4 USDT (Tether)

- Borrowing Asset: USDT
- Collateral: WBTC, WETH, USDC, XTZ

Rationale: Similar to USDC, USDT is a stablecoin, providing a low-risk borrowing option. The collateral can include WBTC, WETH, USDC, and XTZ, with a recommended LTV of around 80% to leverage the stability of the collateral.

7.6.5 XTZ (Tezos)

- Borrowing Asset: XTZ
- Collateral: WBTC, WETH, USDC, USDT

Rationale: XTZ is less commonly used in lending but can be a viable borrowing asset given its staking capabilities. However, due to its volatility, it is advisable to maintain a conservative LTV of 50 – 60% when using WBTC, WETH, USDC, and USDT as collateral.

7.7 Liquidation Threshold

The liquidation threshold is the percentage at which a position is defined as undercollateralized.

7.7.1 WBTC as Borrowed Asset

Conservative Statement: The liquidation threshold for WBTC is set at 60%, which ensures that the collateral value remains significantly above the borrowed amount, providing a buffer against market volatility. This conservative approach helps protect against adverse price movements in the underlying assets.

Aggressive Statement: In an aggressive strategy, the liquidation threshold for WBTC can be pushed to 80%, allowing for greater borrowing capacity. However, this increases the risk of liquidation during market downturns, necessitating vigilant monitoring of collateral values.

7.7.2 WETH as Borrowed Asset

Conservative Statement: WETH has a liquidation threshold of 60%, which allows for a reasonable margin to absorb price fluctuations. This conservative threshold is beneficial for maintaining stability in a volatile market, especially when using stablecoins as collateral.

Aggressive Statement: With an aggressive approach, the liquidation threshold for WETH can reach 78%. This higher threshold enables more significant borrowing but comes with increased risk, as it may lead to quicker liquidations if market conditions deteriorate.

7.7.3 USDC as Borrowed Asset

Conservative Statement: The liquidation threshold for USDC is set at 75%, providing a solid safety net when using WBTC, WETH, USDT, and XTZ as collateral. This conservative level helps protect against adverse price movements in the underlying assets.

Aggressive Statement: An aggressive strategy allows for a liquidation threshold of 82% for USDC, enhancing borrowing potential. However, this approach carries the risk of rapid liquidation if the market experiences sudden downturns, requiring careful risk management.

7.7.4 USDT as Borrowed Asset

Conservative Statement: USDT's liquidation threshold is also set at 75%, ensuring that the collateral remains robust against market volatility. This conservative threshold is particularly effective when stablecoins are used as collateral.

Aggressive Statement: In an aggressive lending scenario, USDT can have a liquidation threshold of 82%. This higher threshold increases borrowing capacity but also necessitates close monitoring to mitigate the risk of liquidation during adverse market conditions.

7.7.5 XTZ as Borrowed Asset

Conservative Statement: The liquidation threshold for XTZ is conservatively set at 52%, which helps maintain a cushion against price volatility. This cautious approach is vital when using more volatile assets as collateral.

Aggressive Statement: Under aggressive parameters, XTZ can have a liquidation threshold of 65%. This allows for increased borrowing potential but raises the risk of liquidation, particularly in a fluctuating market environment.

7.8 Liquidation Penalty

The liquidation bonus is a fee rendered on the price of assets of the collateral when liquidators purchase them as part of the liquidation of a loan that has passed the liquidation threshold.

If the bonus is too low, it may not be sufficient to incentivize liquidators to act during times of extreme volatility or illiquidity. In such conditions, liquidators might not find it profitable to participate, even with the bonus. This can lead to cascading, systemic effects, as a lack of liquidations can cause a downward spiral. Conversely, if the bonus is set too high, suppliers may end up losing a significant portion of their profits to the liquidators. This could make the lending platform less attractive to suppliers.

7.9 Liquidation Factor

The liquidation protocol fee directs a share of the liquidation bonus to a collector contract from the ecosystem treasury. The average liquidation protocol Fee is at 10% in the DeFi Landscape.

7.10 Reserve Factor

The reserve factor allocates a share of the protocol's interests to a collector contract from the ecosystem treasury. The current average Reserve Factor

Market rate is at 5% in the DeFi Landscape.

There is no historical data for Superlend, due to which we cannot recommend some specific number. We will suggest going ahead with the current average Reserve Factor market rate.

8 Results

Risk Parameter	WBTC	WETH	XTZ	USDC	USDT
Isolation Mode	NO	NO	NO	NO	NO
Enable Borrow	YES	YES	YES	YES	YES
Enable Collateral	YES	YES	YES	YES	YES
E-mode Category	N/A	N/A	N/A	YES	YES
Loan-To-Value	55%	55%	47%	70%	70%
Liquidation Threshold	60%	60%	52%	75%	75%
Liquidation Bonus	7%	5%	5%	5%	5%
Reserve Factor	5%	5%	5%	5%	5%
Liquidation Protocol Fee	10%	10%	10%	10%	10%
Supply Cap	2	43	158.6K	100K	100K
Borrow Cap	1.20	25	95K	70K	70K

Table 1: Conservative Risk Parameters

Risk Parameter	WBTC	WETH	XTZ	USDC	USDT
Isolation Mode	NO	NO	NO	NO	NO
Enable Borrow	YES	YES	YES	YES	YES
Enable Collateral	YES	YES	YES	YES	YES
E-mode Category	N/A	N/A	N/A	YES	YES
Loan-To-Value	75%	73%	60%	77%	77%
Liquidation Threshold	80%	78%	65%	82%	82%
Liquidation Bonus	7%	5%	5%	5%	5%
Reserve Factor	5%	5%	5%	5%	5%
Liquidation Protocol Fee	10%	10%	10%	10%	10%
Supply Cap	5	100	300K	200K	200K
Borrow Cap	3	60	180K	160K	160K

Table 2: Aggressive Risk Parameters

NOTE: The data presented in this report have been based on the Etherlink market data and an estimation of data from other chains. These figures are subject to change as they are drawn from a test environment and are intended to provide an initial reference for analysis.

9 Conclusion

The economic audit of Superlend conducted by Chainrisk has yielded promising results, confirming that the protocol’s design is well-aligned with its strategic objectives. Rigorous analysis, simulations, and extreme value analysis demonstrate that the protocol performs robustly.

Our findings demonstrate Superlend’s resilience to scale and withstand high volatility across various collateral types. The system remained significantly over-collateralized even under extreme historical volatility conditions for assets like WETH, WBTC, USDC, USDT, and XTZ.

However, given the dynamic nature of user and market behavior, ongoing parameter evaluation and adjustment are crucial. Also, Superlend needs to implement a continuous regime of regular audits, incorporate community feedback, and adapt to evolving market conditions to ensure ongoing stability and resilience.

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